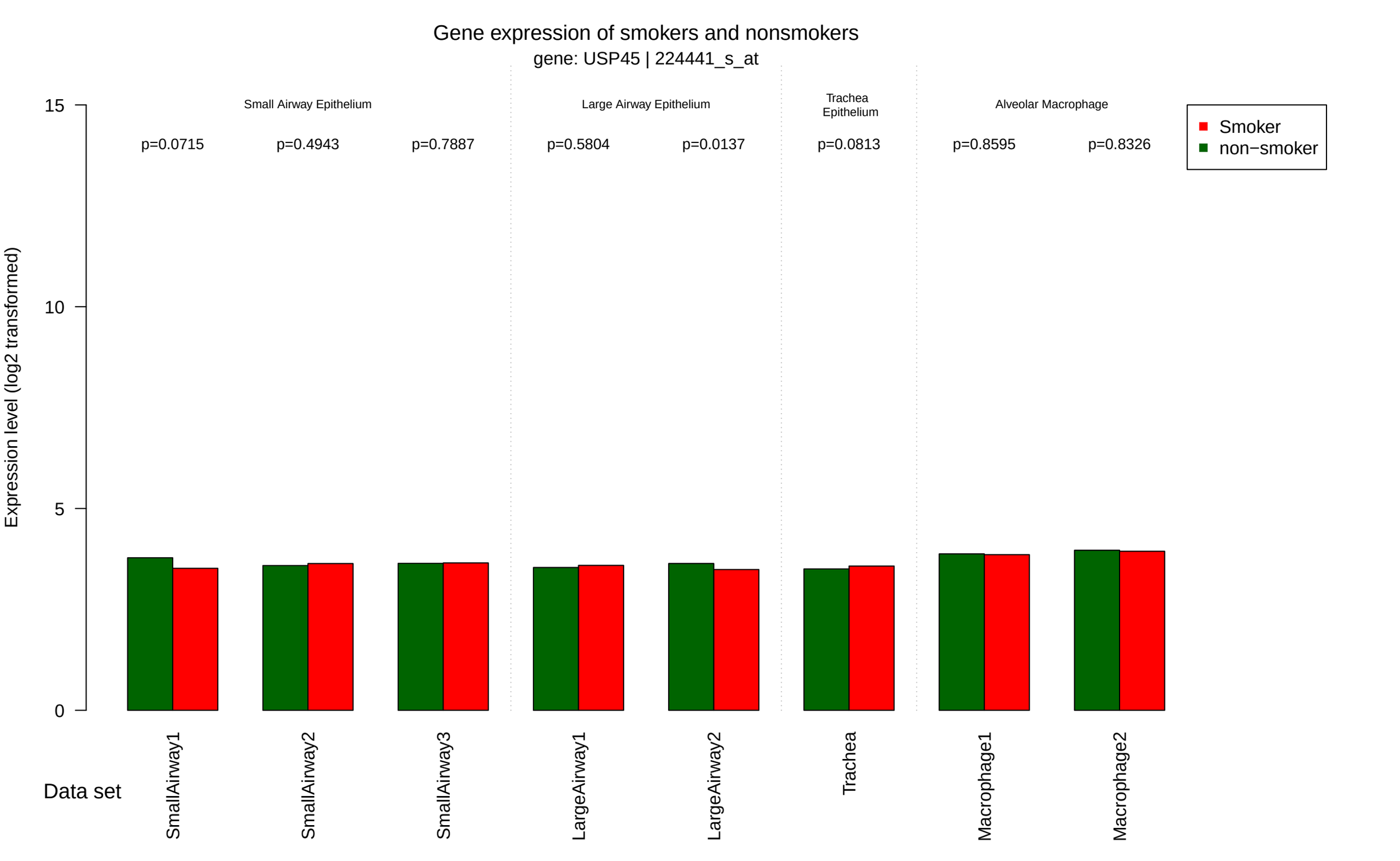
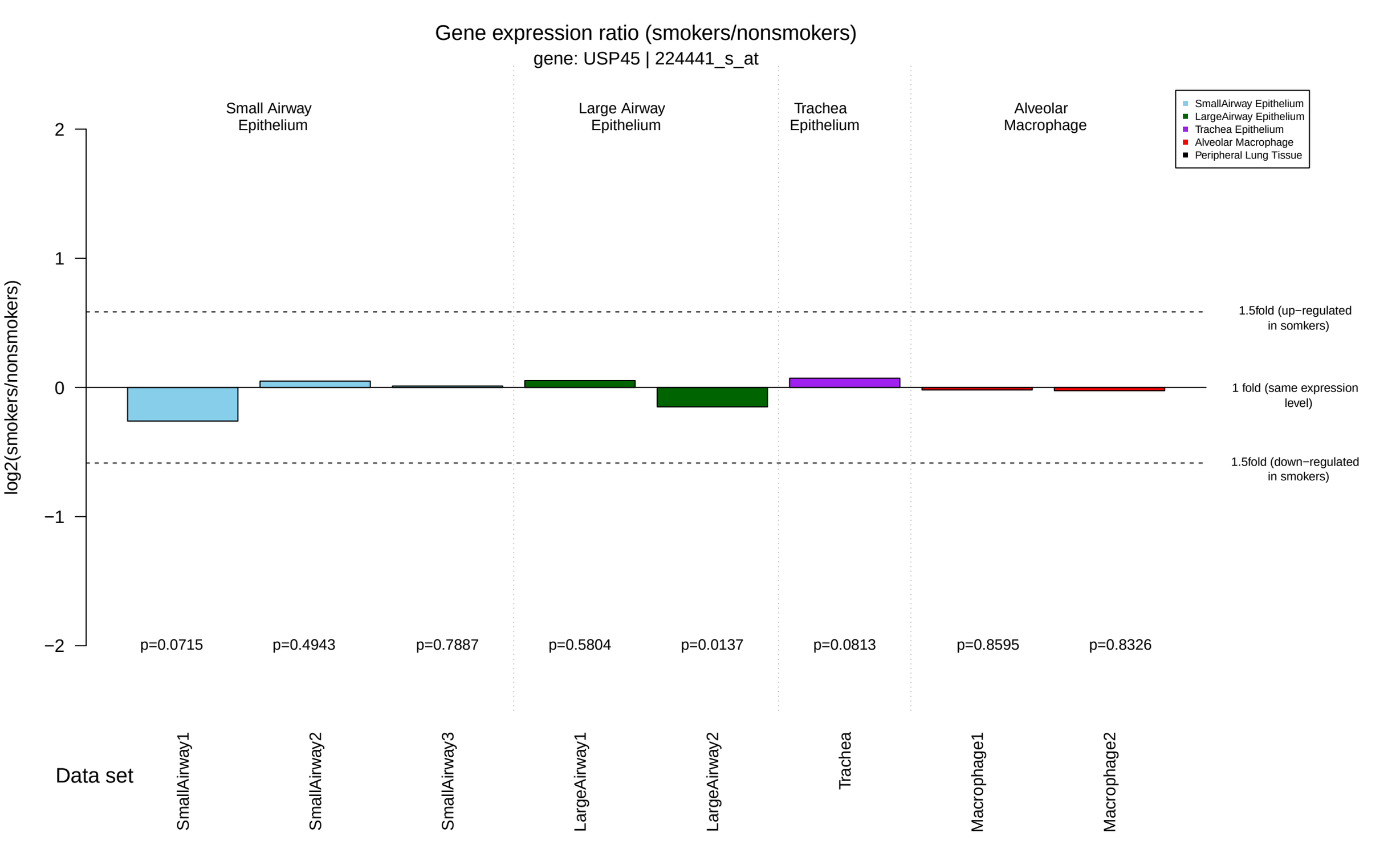
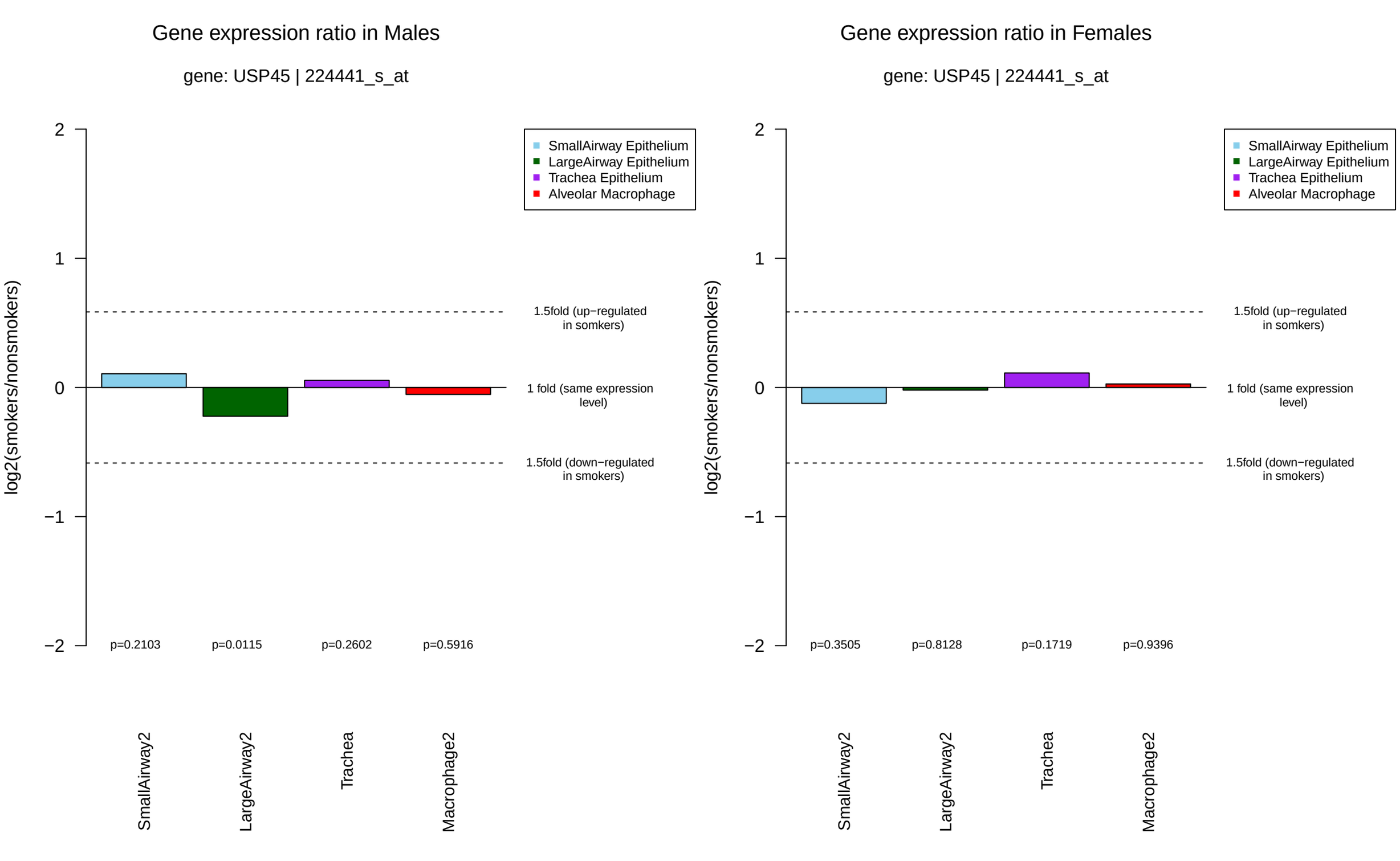
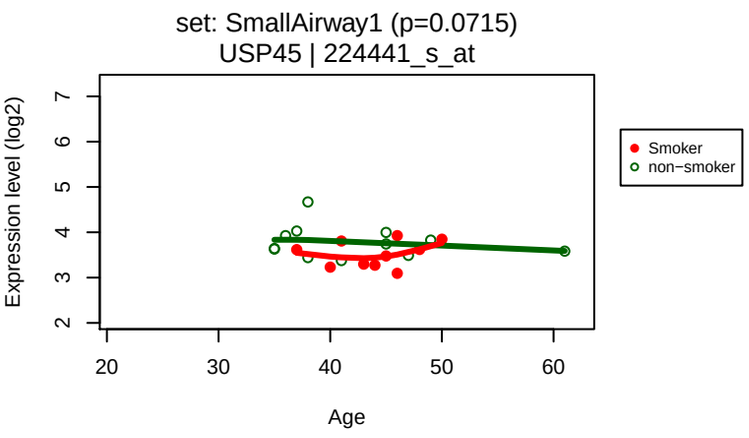
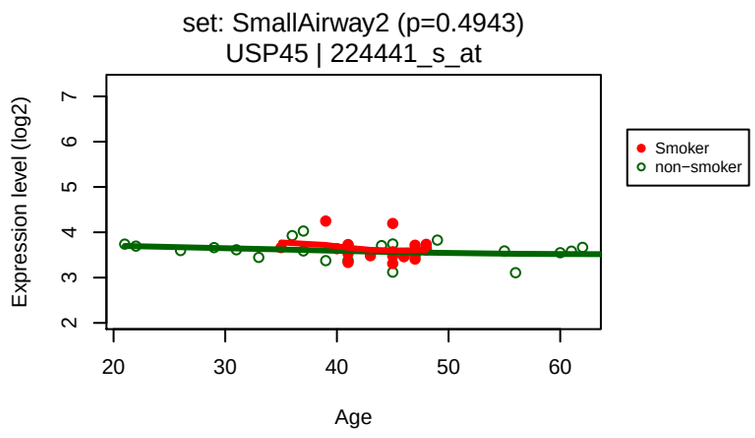
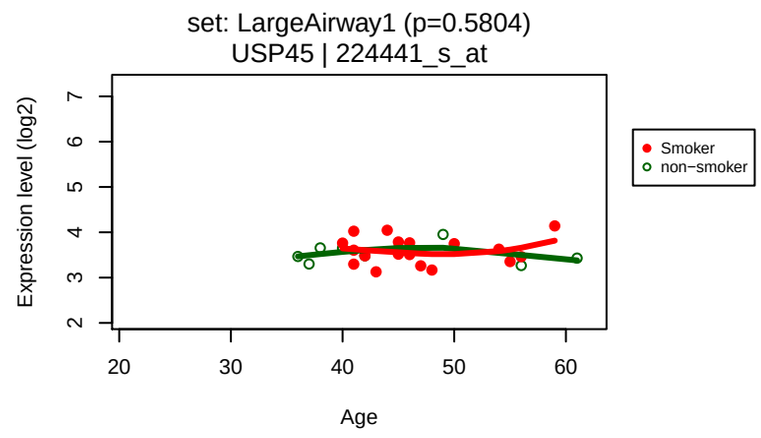

Gene expression of smokers and nonsmokers
gene: USP45 | 224441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0715
p=0.4943
p=0.7887
p=0.5804
p=0.0137
p=0.0813
p=0.8595
p=0.8326
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

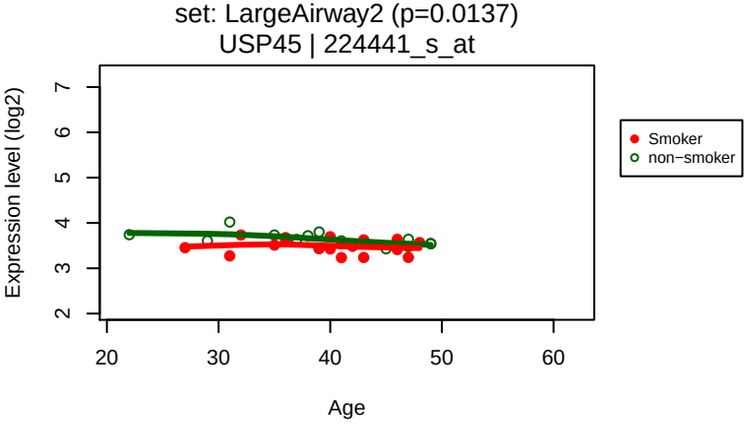

Gene expression ratio (smokers/nonsmokers)
gene: USP45 | 224441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0715
p=0.4943
p=0.7887
p=0.5804
p=0.0137
p=0.0813
p=0.8595
p=0.8326
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

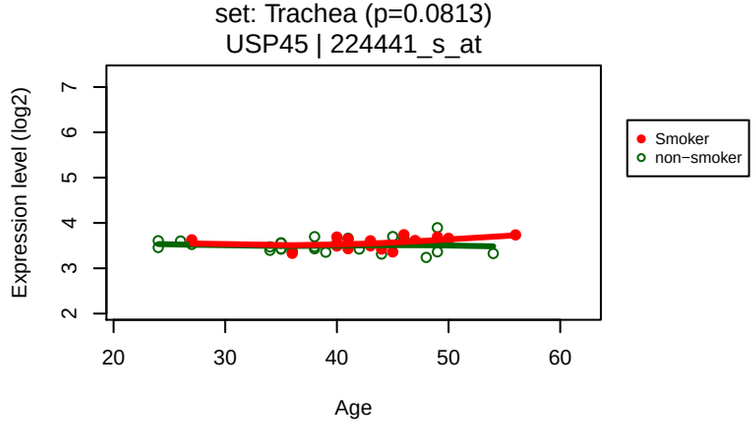

Gene expression ratio in Males
gene: USP45 | 224441_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2103
p=0.0115
p=0.2602
p=0.5916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP45 | 224441_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3505
p=0.8128
p=0.1719
p=0.9396
SmallAirway2
LargeAirway2
Trachea
Macrophage2

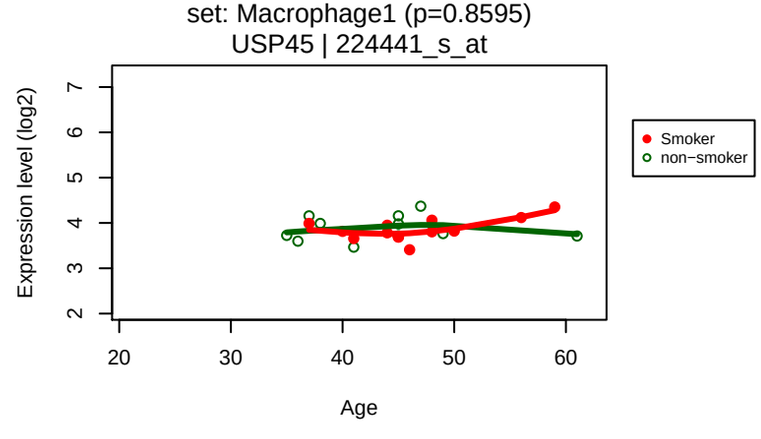

set: SmallAirway1 (p=0.0715)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker

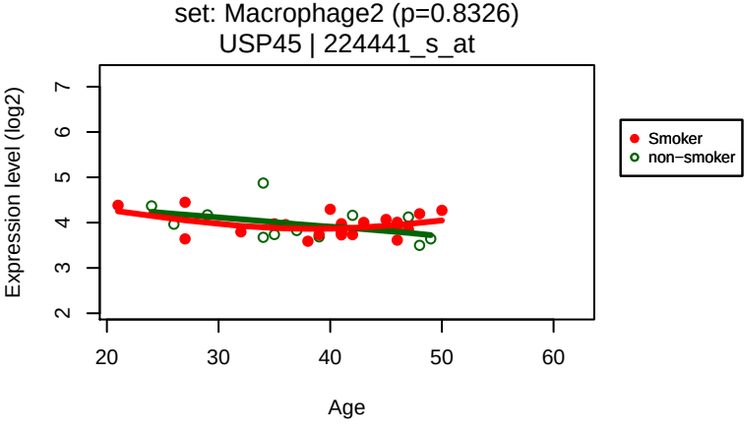

set: SmallAirway2 (p=0.4943)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker

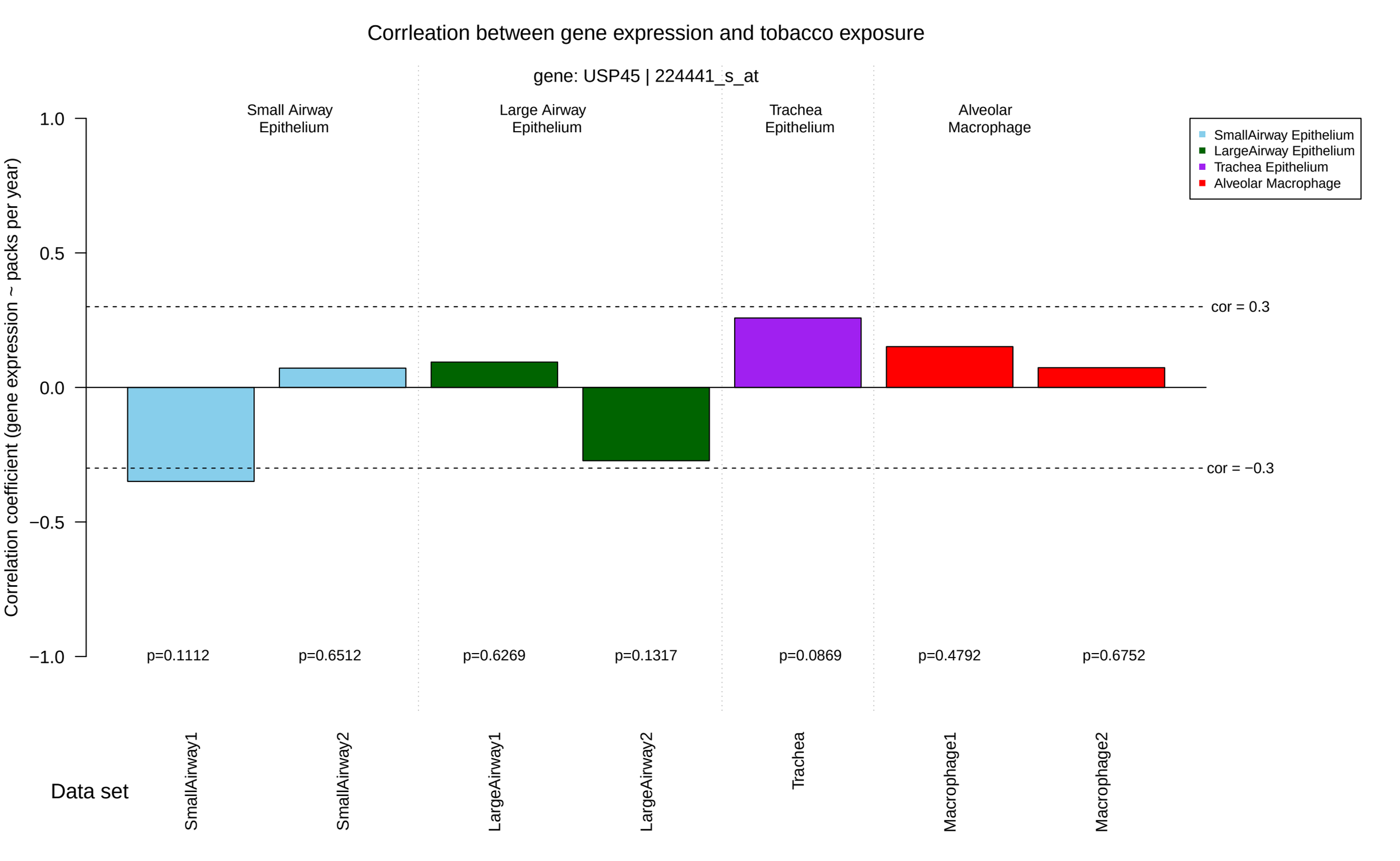

set: LargeAirway1 (p=0.5804)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0137)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0813)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8595)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8326)
USP45 | 224441_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP45 | 224441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1112
p=0.6512
p=0.6269
p=0.1317
p=0.0869
p=0.4792
p=0.6752
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2